

Lumen: An Open, Differentiable, GPU-Parallel Environment for Wall-Safe Endovascular AI

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github.com/SeldingerMed/seldinger-lumen

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Abstract

Endovascular autonomy needs simulation environments that expose the clinical signals a policy must optimize: wall contact, device state, branch selection, image observations, and safety-aware benchmark outcomes. Lumen is an Apache-2.0 endovascular AI environment built around a deformable lumen model, tube-intrinsic contact, GPU-parallel Newton/Warp execution, Gymnasium-compatible navigation tasks, synthetic fluoroscopy, luminal image generation, and replayable computer-vision datasets. The environment is designed to make wall-safe target reaching a first-class benchmark objective rather than an after-the-fact visualization. This preprint describes the system architecture, observation and dataset surfaces, reduced-order flow and device modules, and benchmark semantics used in the public release.

1 Motivation

Autonomous catheter and guidewire navigation has become a credible research direction for robotic endovascular intervention, with reinforcement learning, imitation learning, and inverse reinforcement learning all explored in physical and simulated vessels [3, 4, 1, 5]. CathSim showed that an open simulator can accelerate this work by pairing robot/device simulation with reinforcement-learning tasks [1, 2]. Lumen extends that open benchmark surface toward the signals that are difficult to study in rigid-pipe reaching tasks: deformable-wall interaction, wall penetration, branch-route safety, paired fluoroscopic and state observations, and dataset-grade labels emitted from the same scene.

The design goal is direct: make endovascular navigation trainable as a safety-scored, reproducible AI benchmark. Lumen therefore treats the lumen as a simulated physical substrate rather than only a visual guide. Policies can be evaluated on target reach, route adherence, wall interaction, and image-observation behavior through a shared API.

2 System Overview

Lumen is organized around five connected surfaces:

1. **Procedural vascular cases.** Centerline trees, stenotic segments, tortuous paths, and branch targets are generated as replayable cases. This supports fast benchmark creation and deterministic reruns.
2. **Tube-intrinsic contact and wall state.** Device progress is represented along the vessel manifold, with contact, route progress, wall penetration, torsion, and friction surfaced as environment state.

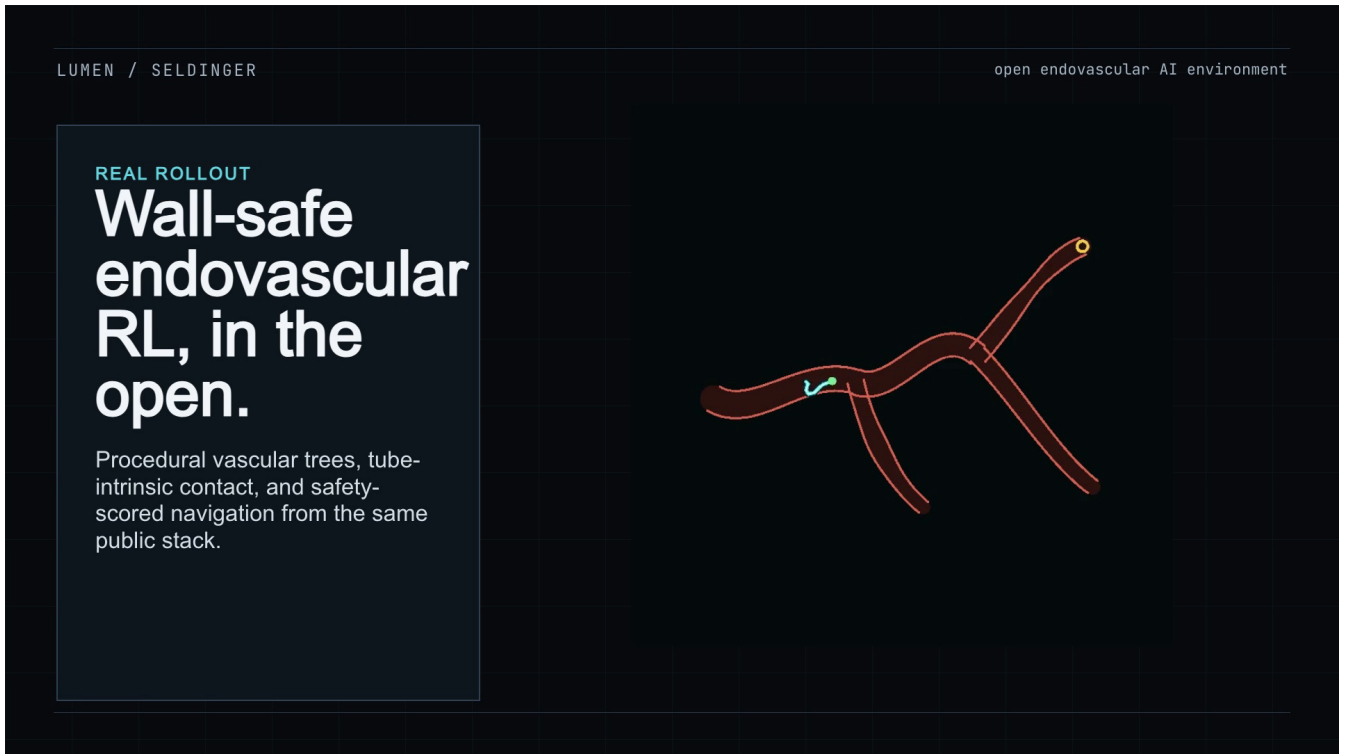


Figure 1: A real Lumen launch capture: a policy rollout in a procedurally generated vascular tree with wall-safety scoring. The launch video and public page use these real simulator-derived captures rather than generated promotional imagery.

3. **GPU-parallel simulation.** The solver path uses NVIDIA Newton/Warp-style kernels for differentiable, high-throughput simulation work [12, 13].
4. **Observation and dataset generation.** Synthetic fluoroscopy, segmentation masks, keypoints, device masks, and luminal RGB views are generated from the same scene graph.
5. **Gymnasium-compatible RL tasks.** Navigation tasks are registered through the Gymnasium interface [15, 14], allowing existing RL tooling to treat Lumen as a standard environment.

This arrangement follows the successful pattern of general RL benchmarks [14, 15], physics-backed control suites [22], and medical-simulation frameworks [6, 7, 8, 9], while specializing the state, observation, and scoring surfaces for endovascular device navigation.

3 Observation Layer

Lumen emits both state observations and image observations. The image side is deliberately multi-modal. A single scene can produce biplanar digitally reconstructed radiographs, masks for vessel and device geometry, labeled keypoints, noisy fluoroscopy, and a luminal RGB camera view. These outputs make the simulator useful not only for reinforcement learning but also for computer-vision pretraining, perception-debugging, and dataset construction.

The dataset workflow is exposed as command-line tooling for capture, validation, indexing, and train/validation/test splitting. Episode sidecars allow users to verify that labels, masks, and replay metadata are present before a dataset is consumed by a model.

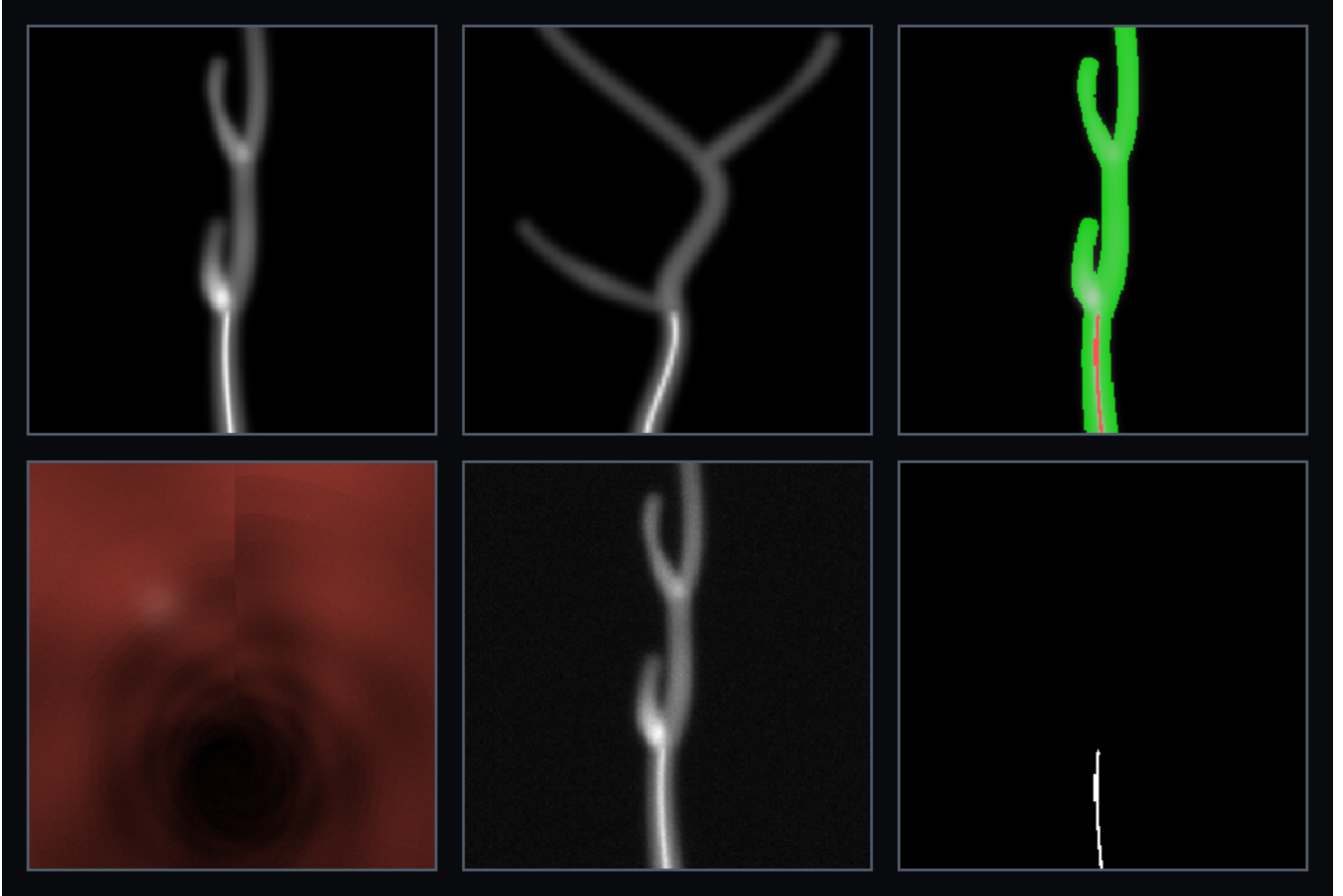


Figure 2: Multi-modal scene outputs from one Lumen case: biplanar fluoroscopy, vessel/device labels, luminal RGB, detector noise, and masks. The same geometry drives control state and image data.

4 Physics and Device Modules

Lumen includes reduced-order modules for the device and vascular phenomena that should influence policy behavior. The release includes:

- deformable-wall semantics inspired by arterial-wall mechanics and anisotropic constitutive modeling [10, 11];
- contact and penetration metrics reported in environment state;
- tortuosity, route progress, and branch-target scoring;
- flow-diverter effects and aneurysm inflow traces;
- clot-field interaction, stentriever retrieval, fragmentation, and damage bookkeeping;
- synthetic fluoroscopy and luminal rendering for image-observation agents.

The simulator is not only a renderer around a centerline. It is a benchmark substrate where contact, flow, clot/device state, and image formation can be captured together, inspected together, and reproduced from case metadata.

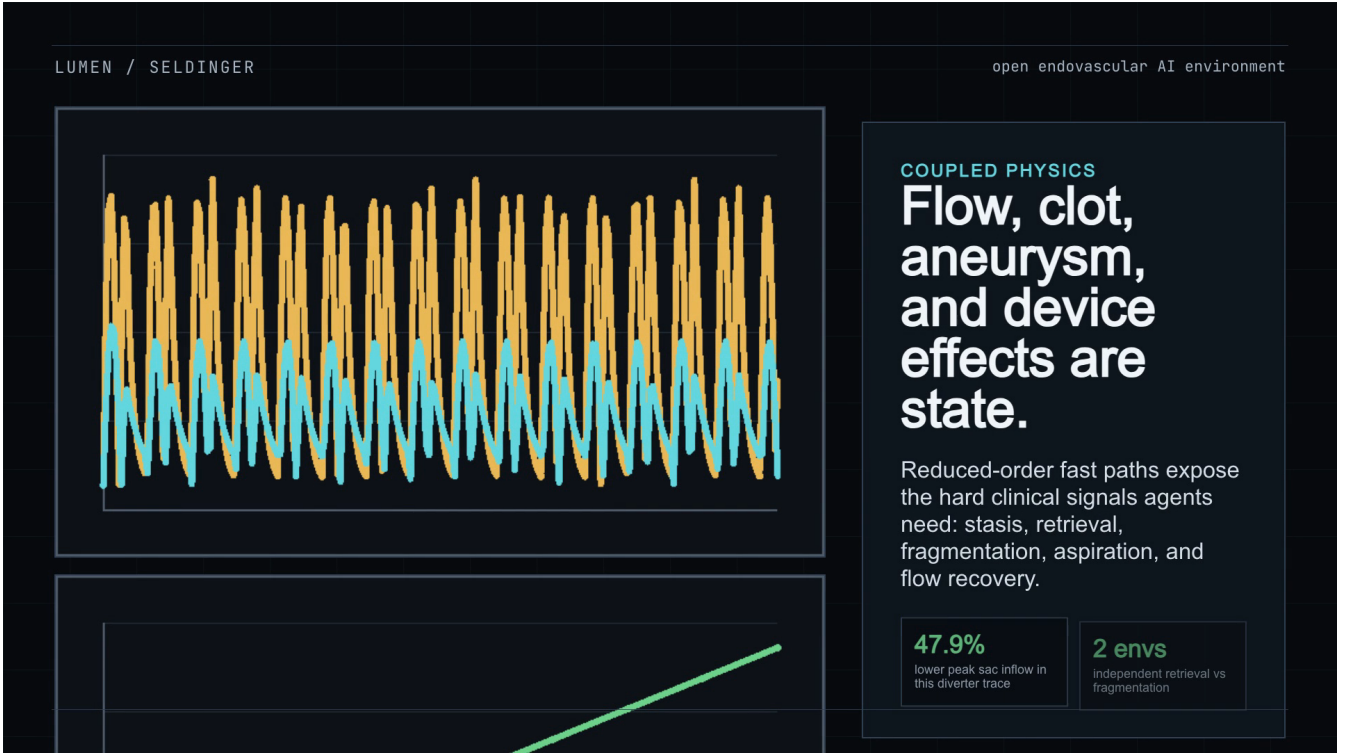


Figure 3: Advanced modules captured from the current implementation: reduced-order aneurysm inflow with a flow-diverter trace and clot/stentriever retrieval state. These modules expose clinically relevant state variables to agents and analysis scripts.

5 Benchmark Semantics

The central benchmark decision in Lumen is to rank *safe success* above raw target reach. A policy that reaches a target while breaching the wall is not equivalent to a policy that reaches the target without unsafe interaction. This distinction matters for endovascular autonomy because the easiest target-reaching policy is not necessarily the most clinically meaningful policy.

Lumen environments can be used with standard reinforcement-learning algorithms such as PPO and SAC [16, 17, 18], while the environment API remains compatible with broader RL infrastructure. The benchmark output is intentionally explicit: route progress, target success, unsafe contact, wall penetration, and replay metadata are all available to downstream evaluation scripts.

6 Positioning Relative to CathSim

CathSim made open-source endovascular simulator research more accessible by providing an autonomous catheterization environment and validating RL tasks in simulation [1, 2]. Lumen is built for the next layer of benchmark pressure: deformable-wall state, wall-safety scoring, multi-modal image labels, replayable dataset tooling, and coupled endovascular state modules in one public stack.

7 Availability

The public repository is:

<https://github.com/SeldingerMed/seldinger-lumen>



Figure 4: Navigation rollouts in branching and tortuous vessels. Lumen reports route progress, target reach, wall penetration, and safety status so benchmark results can distinguish safe target reach from unsafe reach.

The launch page, screenshots, demo video, and this preprint are published at:

<https://seldingermed.github.io/seldinger-lumen/>

8 Conclusion

Lumen provides an open endovascular AI environment where vessel navigation, wall safety, synthetic imaging, labels, and replayable benchmark artifacts are emitted from the same stack. The practical result is a stronger public substrate for studying policies that must do more than reach a coordinate: they must navigate vascular anatomy while preserving wall safety and producing interpretable image-state behavior.

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Table 1: Launch-release surface of Lumen.

Surface	Lumen release behavior
RL interface	Gymnasium-registered navigation tasks and command-line benchmark runs.
Safety scoring	Target reach separated from wall-safe target reach.
Geometry	Procedural stenotic, tortuous, and branching vascular cases.
Physics state	Tube-intrinsic contact, route progress, wall penetration, friction/torsion hooks.
Image observations	Fluoro, masks, keypoints, labels, noisy detector output, luminal RGB.
Dataset tooling	Capture, validate, index, split, and materialize replayable episodes.
License	Apache-2.0 public repository.

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